

Note: 1. Question paper consists of two parts (**Part A and Part B**)

2. Answer all sub questions from **Part A**

3. Answer all questions from **Part B** with **either or choice**

PART - A			10 X 2 = 20 M		
			CO	BTL	Marks
I.	a)	State law of natural Growth.	CO1	L1	2 M
	b)	Find integrating factor of linear differential equation $\frac{dy}{dx} + 2xy = e^{-x^2}$.	CO1	L1	2 M
	c)	Solve $4y''' + 4y'' + y' = 0$.	CO2	L3	2 M
	d)	Find Particular Integral of $(D^2 + 5D + 6)y = e^x$.	CO2	L1	2 M
	e)	Form the partial differential equation by eliminating the arbitrary constants a and b from $z = ax^3 + by^3$.	CO3	L2	2 M
	f)	Solve $px + qy = z$.	CO3	L3	2 M
	g)	Define Gradient of a scalar point function .	CO4	L1	2 M
	h)	Find $\text{div } \vec{r}$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.	CO4	L1	2 M
	i)	Evaluate $\int y^2 dx - 2x^2 dy$ along the parabola $y = x^2$ from (0,0) to (2,4).	CO5	L2	2 M
	j)	State Stoke's theorem.	CO5	L1	2 M
PART - B			5 X 10 = 50 M		
UNIT - I					
1.	a)	Solve $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$.	CO1	L3	5 M
	b)	A body originally at 80°C cools down to 60°C in 20 min. the temperature of the air being 40°C.what will be the temperature of the body after 40 min.	CO1	L3	5 M
(OR)					
2.	a)	Solve $y^2 dx + (-y^2 + x^2 - xy)dy = 0$.	CO1	L3	5 M
	b)	A bacterial culture growing exponentially increases from 100 to 400 grams in 10 hours how much was present after 3 hours ?	CO1	L3	5 M
UNIT - II					
3.	a)	Solve $(D^3 - 6D^2 + 11D - 6)y = e^{-2x} + e^{-3x}$.	CO2	L3	5 M
	b)	Solve $\frac{d^2y}{dx^2} - y = x \sin x$.	CO2	L3	5 M
(OR)					
4.	a)	Solve $(D^2 - 1)y = x^3$	CO2	L3	5 M
	b)	Solve $\frac{d^2y}{dx^2} + y = \sec^2 x$ by the Method of Variation of Parameters.	CO2	L3	5 M
UNIT - III					
5.	a)	Form the Partial Differential Equation by eliminating arbitrary function f from $f(x^2 + y^2, z - xy) = 0$.	CO3	L2	5 M
	b)	Solve $x(y - z)p + y(z - x)q = z(x - y)$.	CO3	L3	5 M
(OR)					
6.	a)	Solve $(D^2 + 4DD' - 5D'^2)z = \sin(2x + 3y)$.	CO3	L3	5 M
	b)	Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^3 z}{\partial x \partial y} + \frac{\partial z}{\partial y} - z = e^{-x}$.	CO3	L3	5 M

UNIT – IV					
7.	a)	Find a unit normal vector to the surface $x^2 + y^2 + 2z^2 = 26$ at the point (2,2,3).	CO4	L1	5 M
	b)	Find the directional derivative of $\phi = xy^3 + yz^3$ at the point (2,-1,1) in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$.	CO4	L1	5 M
(OR)					
8.	a)	Prove that $\vec{A} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational. Find a scalar function f such that $\vec{A} = \nabla f$	CO4	L2	5 M
	b)	Show that $\nabla^2(r^n) = n(n+1)r^{n-2}$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = \vec{r} $	CO4	L2	5 M
UNIT – V					
9.	a)	If $\vec{F} = 3xy\vec{i} - y^2\vec{j}$ then evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the arc of the parabola $y=2x^2$ From (0, 0) to (1, 2).	CO5	L1	5 M
	b)	Find the work done in moving particle in the force field $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the straight line from (0, 0, 0) to (2, 1, 3)	CO5	L3	5 M
(OR)					
10.		Verify Green's theorem for $\int [(xy + y^2)dx + x^2dy]$, where c is bounded by $y = x$ and $y = x^2$	CO5	L2	10 M

*** End of the Question Paper ***

B. Tech II Semester Regular Examinations
DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS
(Common to all Branches)
Model Paper-2

Time: 3 hours**Max Marks: 70M**

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART – A			10 X 2 = 20 M		
			CO	BTL	Marks
I.	a)	State Newton’s law of cooling.	CO1	L1	2 M
	b)	Solve $xdx + ydy = \frac{a^2(xdy-ydx)}{x^2+y^2}$	CO1	L1	2 M
	c)	Find Particular Integral of $(D^2 + 1)y = \cos(2x - 1)$	CO2	L3	2 M
	d)	Solve $(D^3 + 4D^2 + 4D + 4)y = 0$	CO2	L1	2 M
	e)	Form the partial differential equation by eliminating the arbitrary function from $z = f(x^2 + y^2)$.	CO3	L2	2 M
	f)	Solve $4r + 12s + 9t = 0$	CO3	L3	2 M
	g)	Define Divergence of a vector $\vec{f}$	CO4	L1	2 M
	h)	If $f(x,y,z) = 2xz^4 - x^2y$ find $ \nabla f $ at(2,-2,-1)	CO4	L1	2 M
	i)	Evaluate $\oint_C xdy - ydx$ where C is defined by $x^2 + y^2 = 1$	CO5	L2	2 M
	j)	State Gauss divergence theorem	CO5	L1	2 M
PART – B			5 X 10 = 50 M		
UNIT - I					
1.	a)	Solve $\frac{dy}{dx} + x\sin 2y = x^3\cos^2 y$	CO1	L3	5 M
	b)	If the temperature of a cup of coffee is 92 ^o c when freshly poured in a room having temperature 24 ^o c, in one minute it was cooled to 80 ^o c.How long a period must elapse before the temperature of the cup becomes 65 ^o c?	CO1	L3	5 M
(OR)					
2.	a)	Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$	CO1	L3	5 M
	b)	If radioactive Carbon-14 has a half-life of 5750 years, what will remain of one-gram after 3000 years?	CO1	L3	5 M
UNIT – II					
3.	a)	Solve $(D^2 - 4D - 5)y = \cos 2x$	CO2	L3	5 M
	b)	Solve $\frac{d^4y}{dx^4} + \frac{d^2y}{dx^2} = x^3$	CO2	L3	5 M
(OR)					
4.		Solve $(D^2 + a^2)y = \tan ax$ by Method of Variation of Parameters	CO2	L3	10 M
UNIT – III					
5.	a)	Form the partial differential equation by eliminating the arbitrary constants from $z = a \log(\frac{b(y-1)}{1-x})$	CO3	L2	5 M

	b)	Solve $(mz - ny)p + (nx - lz)q = ly - mx$	CO3	L3	5 M
(OR)					
6.	a)	Solve $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^3 z}{\partial x^2 \partial y} + 4 \frac{\partial^3 z}{\partial y^3} = e^{x+2y}$	CO3	L2	5 M
	b)	Solve $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$	CO3	L3	5 M
UNIT - IV					
7.	a)	Find the values of the constants 'a' and 'b' so that the surface $ax^2 - byz = (a + 2)x$ is orthogonal to the surface $4x^2y + z^3 = 4$ at the point (1, -1, 2).	CO4	L1	5 M
	b)	Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at the point P (1, 2, 3) in the direction of the line PQ where Q is the point (5, 0, 4).	CO4	L1	5 M
(OR)					
8.	a)	Find the values of a, b and c such that $\vec{A} = (x + y + az)\vec{i} + (bx + 2y - z)\vec{j} + (-x + cy + 2z)\vec{k}$ is irrotational. Also find ϕ such that $\vec{A} = \nabla\phi$	CO4	L2	5 M
	b)	P.T $\text{div}(r^n \vec{r}) = (n + 3)r^n$. Hence S.T $r^n \vec{r}$ is solenoidal if $n + 3 = 0$ where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = \vec{r} $	CO4	L2	5 M
UNIT - V					
9.	a)	If $\vec{F} = (3x^2 + 6y)\vec{i} - 14yz\vec{j} + 20xz^2\vec{k}$ evaluate line integral from (0,0,0) to (1,1,1) along the path $x = t, y = t^2, z = t^3$.	CO5	L1	5 M
	b)	Find the work done by the force $\vec{F} = (3x^2 - 6yz)\vec{i} + (2y + 3xz)\vec{j} + (1 - 4xyz^2)\vec{k}$ in moving a particle from the point (0,0,0) to the point (1,1,1) along the curve $x=t, y=t^2, z=t^3$	CO5	L3	5 M
(OR)					
10.		Verify Green's theorem for $\int [(3x - 8y^2)dx + (4y - 6xy)dy]$, where c is boundary of the region bounded by $x = 0, y = 0, x + y = 1$.	CO5	L2	10 M

***** End of the Question Paper *****